

TRAVELLING SALESMAN PROBLEM (TSP)

AIM

To develop a Python program that solves the Travelling Salesman Problem (TSP) using the Brute Force approach and finds the minimum-cost tour that visits each city exactly once and returns to the starting city.

OBJECTIVE

- To understand the Travelling Salesman Problem (TSP) in Artificial Intelligence.
- To generate all possible routes (permutations) of the cities.
- To calculate the total cost of each route.
- To determine the minimum-cost tour.
- To display the optimal path and its minimum travel cost.

ALGORITHM

Step 1: Start.

Step 2: Input the number of cities and the cost (distance) matrix.

Step 3: Select one city as the starting city.

Step 4: Generate all possible permutations of the remaining cities.

Step 5: Calculate the total travel cost for each possible tour.

Step 6: Compare the current tour cost with the minimum cost.

Step 7: If the current cost is smaller, update the minimum cost and optimal path.

Step 8: Repeat Steps 5–7 until all possible tours are checked.

Step 9: Display the minimum cost and optimal tour.

Step 10: Stop.

Python Code

```
from itertools import permutations
```

```
graph = [
```

```
[0, 10, 15, 20],  
[10, 0, 35, 25],  
[15, 35, 0, 30],  
[20, 25, 30, 0]  
]
```

```
n = len(graph)  
cities = list(range(n))
```

```
min_cost = float('inf')  
best_path = []
```

```
for path in permutations(cities[1:]):  
    tour = [0] + list(path) + [0]
```

```
    cost = 0  
    for i in range(len(tour) - 1):  
        cost += graph[tour[i]][tour[i + 1]]
```

```
    if cost < min_cost:  
        min_cost = cost  
        best_path = tour
```

```
print("Minimum Cost:", min_cost)  
print("Optimal Tour:", best_path)
```

OUTPUT

```
Minimum Cost: 80
Optimal Tour: [0, 1, 3, 2, 0]
|
```

RESULT

The Python program successfully solves the Travelling Salesman Problem using the Brute Force approach and finds the optimal tour with the minimum travel cost.